



Hilbert's Programme and Georg Kreisel I: The Post-Gödelian Perspective

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Received: 5 February 2025 / Accepted: 10 November 2025 / Published online: 18 February 2026
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Abstract

We discuss Georg Kreisel's perception of Hilbert's Programme, paying particular attention to his *post-Gödelian perspective*.

Keywords David Hilbert · Georg Kreisel · Hilbert's Programme · Gödel's Incompleteness Theorems · Philosophy of Mathematics

"Hilbert programme: a serious project"

Georg Kreisel (1970, p. 32).

1 Introduction

Georg Kreisel (1923–2015) was an eminent logician who significantly contributed to the development of modern logic during the second half of the 20th century. At the same time he had a unique personality which made him famous as a provoking and sometimes insulting character. He enjoyed questioning the rationale of logical research, particularly in proof theory as it emerged from *Hilbert's programme*.

In this paper we will discuss Kreisel's criticism of Hilbert's Programme, focusing particularly on his reservations about its philosophical rationale. We will make extensive use of citations from Kreisel's writings. One problem with Kreisel's criticisms is that they are usually not immediately supported by clear arguments. Instead, they stem from a cloud-like collection of technical results and non-technical

reflections scattered across various papers.¹ For the generation of logicians who knew Kreisel personally, his writings were complemented by oral conversations and an extensive correspondence which probably reinforced the criticisms discussed here.² Later generations can only get an vague idea of his personality by reading the various recollections in the *Kreiseliana* (Odifreddi 1996). In addition, one can deduce conclusions from his numerous and sometimes infamous reviews.

It seems to be common knowledge that Kreisel was highly critical of the philosophical aims of Hilbert's Programme, but searched for mathematically fruitful alternatives based on the technical results developed for this Programme. The textual evidence, however, shows that Kreisel's philosophical evaluation of Hilbert's Programme is not entirely negative. Where it is negative, one can see that it would be historically misplaced to dismiss Hilbert's original conception of his programme, as it is entirely based on later developments and results. At best, it can be used against naive extrapolations of Hilbert's original ideas.

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¹ Complaints about Kreisel's writing style are frequent. We may just cite Takeuti: "It is often said that it is impossible to read his articles, and that it is not clear what is written in them..." (Yasugi and Passell, 2003, p. 69); and Barwise (1996, p. 21): "The paper had become 'typical Kreisel,' full of long asides, footnotes, convoluted qualifications, and modifiers — all the things which make his articles so impenetrable." Kreisel also revealed his intended audience when he responded to Barendregt's complaint that he could not understand many of his writings (1996, p. 5, footnote 3): "Oh, but Gödel can, and Bernays can."

² The author of this paper met Kreisel three or four times, but never had a personal conversation with him.

2 Hilbert's Programme in context

In response to the paradoxes in set theory, which called into question the traditional security of mathematical reasoning, and the challenges raised by Brouwer's intuitionism, David Hilbert (1862–1943) pursued a foundational programme in the 1920s to secure mathematical reasoning through formal consistency proofs. This programme consists of two stages: first, usual mathematical reasoning needs to be formalized in axiomatic theories; secondly, these axiomatic theories should be proven consistent using *finitist* reasoning.³ With the first step, Hilbert subjected mathematical reasoning to mathematical investigation. The second step was based on the underlying rationale that finitist mathematics has to be accepted as consistent due to its *contentual* (“inhaltliche”) nature.

The incompleteness theorems of Kurt Gödel (1906–1978) show that Hilbert's programme cannot be carried out in its original form. However, this result did not put an end to Hilbert's Programme. Instead it prompted Hilbert to take a *new meta-mathematical standpoint*⁴, as Gödel's result “shows only that—for more advanced consistency proofs—the finitistic standpoint has to be exploited in a manner that is sharper [...]”⁵. In 1936, Gerhard Gentzen (1909–1945) provided a consistency proof of number theory using transfinite induction up to ε_0 , which aligned with the new methodological standpoint and which gave rise to the *Revised Hilbert Programme*. In this revised version, consistency proofs of formalized mathematical theories should be performed using “restrictive means”, but not only finitary ones. Such restricted means are often called “constructive”, but Kreisel pointed out⁶ that this would not be very meaningful—what would be a non-constructive consistency proof anyhow?

In both versions, in the original finitist one as well as in the revised form, Hilbert's programme clearly targets consistency. It is important, however, to remember that this question was embedded in a much larger foundational

research programme of Hilbert which dates to the end of the 19th century. The first published traces of this program can be found in Hilbert's seminal lecture on 23 mathematical problems for the 20th century, presented at the International Congress of Mathematicians in Paris (Hilbert 1900).

The quest for a consistency proof of the “arithmetical axioms”—working with higher-order logic, Hilbert used this term for the axioms of analysis—was prominently placed in the second position. The underlying motivation has two sources: 1. the mathematical concern with the set-theoretic paradoxes, and the resulting question about the reliability of mathematical reasoning, and 2. a rejection of the philosophical pessimism expressed in the dictum *ignoramus et ignorabimus* of Emil du Bois-Reymond (1818–1896), see (Bois-Reymond 1872).

Gerhard Gentzen (1969, p. 237) described Hilbert's conceptional plan as follows: “A foremost characteristic of Hilbert's point of view seems to me to be the endeavour to withdraw the problem of the foundations of mathematics from *philosophy* and to tackle it as far as in any way possible with methods proper to mathematics.”

In his talk *Axiomatisches Denken*, presented at a meeting of the Swiss Mathematical Society in Zurich in 1917, Hilbert provided a lengthy list of objectives for the then-still-to-be-developed field of *proof theory* (Hilbert 1996, p. 1113):

But the completion [of the *axiomatization of logic*] will require further work. When we consider the matter more closely we soon recognize that the question of the consistency of the integers and of sets is not one that stands alone, but that it belongs to a vast domain of difficult epistemological questions which have a specifically mathematical tint: for example (to characterize this domain of questions briefly) the problem of the *solvability in principle of every mathematical question*, the problem of the subsequent *checkability* of the results of a mathematical investigation, the question of a *criterion of simplicity* for mathematical proofs, the question of the relationship between *content and formalism* in mathematics and logic, and finally the problem of the *decidability* of a mathematical question in a finite number of operations.

To address these issues, Hilbert and his school in Göttingen developed *proof theory* as a discipline within the emerging field of mathematical logic in the 1920s. While consistency was certainly a focus of Hilbert's foundational research, it was less important compared to the overall aim of proof theory. Kreisel gave us an impressive testimony of this fact, see Citation 20 below.

³ Kreisel was well aware of this separation, see Citation 19 below.

⁴ Ackermann (1933, p.1f) writes in a letter to Hilbert (August 23rd, 1933): “Besonders interessiert hat mich der neue meta-mathematische Standpunkt, den Sie jetzt einnehmen und der durch die Gödelsche Arbeit veranlaßt worden ist.” (“I was particularly interested in the new meta-mathematical standpoint which you now adopt and which was provoked by Gödel's work.”). And Reid (1970, p. 198) reports of Bernays testimony: “At first [Hilbert] was only angry and frustrated, but then he began to try to deal constructively with the problem. Bernays found himself impressed that even now, at the very end of his career, Hilbert was able to make great changes in his program.”

⁵ Hilbert and Bernays (2011, p. VII). German original: “Jenes Ergebnis zeigt in der Tat auch nur, daß man für die weitergehenden Widerspruchsfreiheitsbeweise den finiten Standpunkt in einer schärferen Weise ausnutzen muß, [...]”

⁶ According to a personal communication of Matthias Baaz (Vienna).

However, even with respect to consistency, one must distinguish between Hilbert's bold hope of eliminating philosophy from mathematics and a much more modest goal. The latter was merely to rebut intuitionism, as it threatened parts of mathematics that Hilbert was unwilling to negotiate. He wanted to force intuitionists to accept the consistency, and *a fortiori*, the meaningfulness of classical mathematics.⁷ In this respect, his programme was based on a modus tollens argument: if (informal) mathematical reasoning were subject to hidden contradictions, then those contradictions would carry over to formalized mathematics. If one could demonstrate that formalized mathematics is free of contradictions, then informal mathematical reasoning would be validated. By carrying out such a consistency proof using reasoning that is acceptable to intuitionists, they would have been beaten at their own game.⁸

3 Kreisel's Criticism of Hilbert's Programme

Kreisel published around 200 papers, many of which contain comments relating to Hilbert's program in one way or another. His contribution to the Festschrift for the 70th birthday of Paul Bernays (1888–1977) stands out in this respect, bearing the title *Hilbert's programme* (Kreisel 1958a).⁹

⁷ In reality, Hilbert and Brouwer were talking at cross-purposes: Brouwer did not even deny the possibility that Hilbert's programme could be carried out. He just insisted that consistency would not imply meaningfulness; see (Brouwer, 1967a, Fourth insight, p. 492. He uses the term *correctness* instead of our *meaningfulness*). He also expressed this by saying (Brouwer 1967b, p. 336): "an incorrect theory, even if it cannot be inhibited by any contradiction that would refute it, is none the less incorrect, just as a criminal policy is none the less criminal even if it cannot be inhibited by any court that would curb it." Hilbert never responded to this allegation—but he had already dismissed it before in his talk "Über das Unendliche" (Hilbert 1926, p. 163), and Brouwer cites this dismissal as evidence of Hilbert's lack of understanding. In any case, the issue must be raised anew after Gödel's results. It was, however, echoed by Bishop (1985, p. 1) when he spoke about "the debasement of meaning" in classical mathematics.

⁸ Kreisel emphasized the observation that Hilbert's finitism is weaker than intuitionism, see (Kreisel 1958a). Hermann Weyl (1885–1955) implicitly expressed this already in his comment that Hilbert was "more papal than the pope" (Weyl 1944, p.641).

⁹ Kreisel did not coin the term. As far as we can tell, it was not used by Hilbert and his collaborator in the 1920s and '30s; Hermann Weyl (1944, p. 644) speaks about a "program" in his obituary for Hilbert, but only briefly and without mentioning Hilbert by name. One can find it, in the way as we use it today, in Kleene's monograph *Introduction to Metamathematics* of 1952 (Kleene (1952), p. v and p. 56). We also found an earlier appearance in print in Reichenbach's *The Rise of Scientific Philosophy* (Reichenbach 1951, p. 226). However, the term gained significant publicity when Kreisel's paper was included (in a slightly extended version, (Kreisel 1964)) in the collection of papers on Philosophy of Mathematics by Benacerraf and Putnam (1964);

At least three different topics can be identified in Kreisel's criticisms:

1. Philosophical and formal reservations concerning Hilbert's original aims;
2. Finitism, intuitionism and constructivism as a basis for a Hilbert programme;
3. Questions concerning decidability.

In this paper, we will focus on the first item, deferring discussion of the second and third items to a later opportunity. As we will argue in §4, Kreisel's criticisms concerning the first item are biased by a *post-Gödelian perspective*.

3.1 The "failure" of Hilbert's Programme

In Kreisel's presentation (1958a, Abstract) of Hilbert's (Finitist) Programme we find explicitly formulated its *failure*. The unfeasibility of Hilbert's original programme was recognized immediately after Gödel's results in Göttingen. This led, as described in §2, to the "modified methodological standpoint" of Hilbert which resulted in the Revised Hilbert Programme. However, Kreisel's harsh wording¹⁰ disqualifies Hilbert's Programme in a quite strong way:

Citation 1 ((Kreisel 1958a, p. 352)) *As far as an overall-philosophical understanding is concerned, the original Hilbert programme has failed, and, as is usual with great schemes, it gives no hint of what might take its place.*

A fair evaluation of Hilbert's Programme would consider it as a mathematical conjecture that has been disproven. However, Kreisel went much further. As Hilbert's declared goal was to remove philosophy from mathematics once and for all, Kreisel could not see how this could be have been achieved if the programme had worked:

Citation 2 ((Kreisel 1970, p. 33)) *Of course some of the uses to which Hilbert wanted to put his programme were ill thought out, for instance, by § 10, even its success would not have provided the separation of mathematics from philosophy which Hilbert wanted.*

This citation is a prime example of the problem mentioned in the introduction regarding Kreisel's writings: it lacks clear supporting arguments. The reference to § 10 seems useless. That paragraph, entitled "Separating mathematical practise from its logical analysis" contains some

in the second edition of this collection, Kreisel (1983) added a postscript and several endnotes to update his views as of autumn 1978.

¹⁰ The German version of the abstract speaks about "Versagen", which is far from being polite.

interesting considerations about such a separation. But there is nothing close enough to Hilbert's original aims to conclude that his programme was ill thought out.

However, one may find the underlying argument in other papers:

Citation 3 ((Kreisel 1968, p. 322)) *Consequences of Gödel's Theorems for proof theory. [...]*

There will be a need for a philosophical analysis of conceptions $\mathcal{P}$ [apparently of a higher mathematical theory, as Arithmetic], other than $\mathcal{P}_0$ [finitist mathematics], and the interest will depend on the intrinsic significance of $\mathcal{P}$. Thus we do not have the separation which Hilbert wanted.

Gödel's Theorems imply the non-conservativity of higher mathematical theories over more basic ones. In this respect, Kreisel is correct in saying that one cannot expect a basic theory, such as finite mathematics in Hilbert's original conception, to epistemologically exhaust all of mathematics.¹¹

Essentially, Kreisel's criticism is directed at the philosophical rationale of Hilbert's (finitist or revised) Programme. In contrast, Kreisel was interested in the technical side of the program, to which he made substantial contributions, both conceptional and technical. In this respect, Howard (1996, p. 276) writes:

The use of a program of this sort in order to *justify* classical mathematics has been severely criticized by Kreisel over the years. Kreisel's criticism is supported by the following phenomenon. As time has gone by, the constructive methods have become more and more abstract and complex. As an example of complexity: it is difficult to understand the ordering of the ordinal notations needed for the treatment of, say, Π_1^1 comprehension. Thus one feels that the constructive principles of [the constructive system] $\mathcal{L}^*$ are no more "understandable" than the subsystem $\mathcal{L}$, of classical mathematics being treated. Kreisel has made repeated attempts to find new aims for proof theory. Aside from the question of the philosophical issues, these attempts have had a beneficial effect on the purely technical side of proof theory.

3.2 Positive evaluation of the programme

Despite questioning the philosophical rationale of Hilbert's programme in its original form, Kreisel offers a distinct

positive evaluation of the programme with regard to the following issues:

1. Providing the necessary foundation for its refutation;
2. Providing tools to obtain interesting mathematical results, in particular, concerning constructive aspects;
3. Partial realizations.

In addition, he also stresses that

4. Hilbert's programme must be sharply distinguished from naive (or "crude") formalism.

3.2.1 Hilbert as preparation for Gödel

Kreisel states in several places that Hilbert's programme led to precise progress because it was a necessary step to reach Gödel's results:

Citation 4 (Citation 2 extended) *The single most important thing to remember about Hilbert's programme (in contrast to naive formalism) is this : it was not an empty shell, it was something that led to precise progress. It (or rather the assumption behind it [...]) was refuted by a splendid theorem, not by the kind of wise cracks which naive formalism invites. Of course some of the uses to which Hilbert wanted to put his programme were ill thought out, [...], even its success would not have provided the separation of mathematics from philosophy which Hilbert wanted. But, against this, for some areas of informal mathematics which are about abstract objects, Hilbert's programme can be carried out with striking results.*

On another occasion, he was more terse:

Citation 5 ((Kreisel 1976b, p. 179)) *[...] the work on Hilbert's Programme does not turn out to be useless: without it, the assumptions, which led to that programme,—in my view—would never have been convincingly refuted.*

The appreciation of Hilbert even culminated in the following comment on Gödel's work:¹²

¹² This citation is embedded in a note in which Kreisel recalls his satisfaction with the success of a provocation presented in his obituary for Gödel (Kreisel 1980):

Citation 6a (Citation 6 extended) When writing this article I had [(Kreisel 1980)] very much in mind, and some reactions by the — absolutely, very small — audience to which it was addressed. [...] On the positive side, at least for *all* the relatively many readers who commented on the matter to me, the most unconventional twist of [(Kreisel 1980)] had precisely the desired effect. (I mean of course the introductory fanfare about Gödel not having invented mathematical logic.) All those congenial to me — told me that they — looked immediately at

¹¹ It remains unclear, however, why the "intrinsic significance" of higher mathematical concepts should not be accessible to higher proof-theoretic analysis. Kreisel's skepticism appears to be based (only?) on the difficulties explained in the following citation of Howard.

Citation 6 ((Kreisel 1987, p. 177)) *[T]he problem [Gödel] solved had been formulated and made famous, mainly by Hilbert!*

3.2.2 Other mathematical results

Kreisel did not see the value of Hilbert's programme in any kind of consistency proof, which was supposed to have philosophical significance. Rather, he saw the value in the possibility of using such consistency proofs (or of the techniques developed for them) to obtain *other* results of *mathematical* interest.

Citation 7 ((Kreisel 1958a, p. 351)) *In my opinion, Hilbert's programme [...] is a rich line of research in foundations. [...] Also, the (modified) Hilbert programme gives scope to a great variety of methods of mathematical logic including those of the topological approach, intuitionism, recursion theory [...].*

As far as piece-meal-understanding is concerned, its importance consists in having let to the fruitful study of the constructive aspects of axiomatic systems. [...] My own interest in the modified Hilbert programme does not go one way only, i.e. the elimination of non-constructive methods, but I find that greater facility with the non-constructive methods comes from a study of their constructive aspects.

Only after this positive evaluation did Kreisel continue with the philosophical criticism given in Citations 1.

On another occasion, he combined his appreciation with a rather mild criticism of Hilbert's programme:

Citation 8 ((Kreisel 1958b, p. 155)) *The avowed purpose of all this work [of the Hilbert school in the foundations of mathematics] is described in the introduction to volume I of Hilbert-Bernays: the consistency of the usual principles of proof is to be established by means of finitist methods, or at least, by means of methods which are more "evident" or more "constructive" than the principles under discussion. This formulation seems to have several defects:*

- (1) *Since the notion of constructive proof is vague, the whole formulation of the program is vague; and though an exact formulation constitutes, of course, an interesting problem for the logician, because of this vagueness the mathematician does not find the program attractive.*

the final paragraph for the so to speak compensating twist; all those of a different species [...] — told me they — were ill at ease about the fanfare or even misinterpreted it by overlooking the fact that *all* of Gödel's famous work in the thirties had been prepared for him; the problems he solved had been formulated and *made* famous, mainly by Hilbert!

- (2) *The formulation does not cover too well the actual substance of the material contained in [(Hilbert and Bernays 1934)]; e.g. the ϵ -theorems for the predicate calculus go far beyond establishing mere consistency of the predicate calculus. [...]*

There is a different general program which does not seem to suffer the defects of the consistency program: To determine the constructive (recursive) content or the constructive equivalent of the non-constructive concepts and theorems used in mathematics, particularly arithmetic and analysis.

He also reiterated this point in continuation of Citation 5:

Citation 9 ((Kreisel 1976b, p. 179)) *But the correction (of naive assumption) exhaust not yet the interest in this work, in particular of Gentzen's consistency proof. One has to look newly to the consistency proof and to use the ideas of the consistency proof for proofs of other results [...]*

Bernays (1978, p. 22) even approved this "shift of emphasis":

Proof Theory pursued a different path than Hilbert had imagined. [...] The significance of such [consistency] proofs lies not really in generating convincement of the consistency—in particular KREISEL pointed this out—; instead in each case such a proof shows something more advanced than just the consistency.

Kreisel's most important legacy is probably promoting this fruitful perspective for research in proof theory. See also Kohlenbach's evaluation in (2023).¹³

3.2.3 Partial realizations

As we recalled in Citation 4, Kreisel acknowledged that "for *some* areas of informal mathematics which are about abstract objects, Hilbert's programme can be carried out with striking results."

In another paper, Kreisel highlights the (initial) success of Hilbert's programme:¹⁴

Citation 10 ((Kreisel 1976b, p. 204)) *One has to stress that the early stage of the theory [proof theory] was decidedly satisfactory and that the results of that time remain obviously of interest; in particular, (i) Hilbert's mathematical clarification of a widespread impression concerning the role of formal rules in Mathematics, (ii) the proof—by*

¹³ This paper also contains some more citations of Kreisel, which complement the ones given here.

¹⁴ The third item reiterates the value of Hilbert's programme to arrive at Gödel's results.

appropriate formalization—that the impression (i) for many areas of (then) mathematical practise is correct, and finally (iii) the theoretical refutation of (i) by Gödel’s incompleteness theorems.

Even though this appreciation comes—once again—with a slight negative twist at its end, it shows that Kreisel was able to place Hilbert’s Programme in the correct historical context.

3.2.4 Naive formalism

In line with the preceding arguments, he calls Hilbert’s programme “a serious project”—rejecting only the “naive formalism” which others supposedly derived from it:

Citation 11 ((Kreisel 1970, p. 32)) *Aware of the weakness of naive formalism, Hilbert made a serious coherent modification; more precisely, a conjecture which should justify the autonomy of a genuine part of mathematics, containing general assertions, not only mere computation as in naive formalism.*

More than 50 years after Frege’s criticisms, Gödel’s incompleteness theorem appeared and made Hilbert’s conjecture quite implausible, theoretically speaking. Practically, it was as good as abandoned. People were of course familiar with the programme with its emphasis on formalization, they were not aware of Frege’s criticism, and, as people will, they kept the trash of naive formalism in Hilbert’s version. [...]

13. Hilbert programme: a serious project...

And Hilbert himself is correctly cleared of any accusation of naive or crude formalism.

Citation 12 ((Kreisel 1964, p. 158)) *Hilbert was certainly not a fanatic of crude formalism: thus while his problems concerned syntactic properties of formal systems, the solutions were to be given by intuitively correct reasoning, and he explicitly considered any formalization of this reasoning unnecessary.*

It remains unclear whom Kreisel has in mind as a “crude formalist”, but he could have been referring to Bourbaki. At least he states that “Bourbaki flirts with this doctrine” (Kreisel and Krivine (1967), p. 221, footnote 4).

3.3 Consistency

In sharp contrast to the positive evaluations given in the previous section, Kreisel repeatedly criticized Hilbert’s interest in consistency, even calling it *logical atrocities*:

Citation 13 ((Kreisel 1987, p. 78)) *Looking back at the early thirties, when Gödel did his best known work on i[ntuitionistic] l[ogic], the first order of business was to help clean up the logical pollution spread by Brouwer and his epigones; comparable to Hilbert’s logical atrocities with his claims for consistency...*

And he admits his personal antipathy:

Citation 14 ((Kreisel 1987, p. 148)) *[Gödel] knew very well my aversion to Hilbert’s claims for consistency;*

The main argument for this aversion seems to be the following:

Citation 15 ((Kreisel 1976b, p. 179)) *The doubts about the consistency of Arithmetic [...] are in turn indeed more problematic than the consistency itself.*

This argument was repeated at various places and was not only restricted to Arithmetic.

In contrast, Kreisel noted the importance of developments in set theory in increasing the mathematician’s confidence:

Citation 16 ((Kreisel 1983, p. 231, *Postscript (Autumn 1978)*)) *[T]he clear description of segments of the cumulative hierarchy of sets has done more for the removal of genuine doubts than all finitist consistency proofs put together.*

3.4 Gödel’s provability predicate

In technical presentations, Kreisel preferred to restate Hilbert’s programme in modern formal terms. In particular, he used Gödelian proof and provability predicates.

Citation 17 ((Kreisel 1965, p. 151)) *Hilbert’s basic considerations were (more or less) these: [...]*

In consequence we have an arithmetic relation $\text{Prov}_F(n, \ulcorner A \urcorner)$: n is a formal derivation by the rules F of the formula A .

One page later, this is used to claim:

Citation 18 ((Kreisel 1965, p. 152)) *Hilbert also emphasized the amusing fact that [...] 3.1133 $[\text{Prov}_F(m, \ulcorner P(x) \urcorner) \Rightarrow P(n)]$ is an immediate corollary of the consistency statement (with variable formula A), variables m and n over (numbers of) formal proofs:*

$$\text{Prov}_F(m, \ulcorner A \urcorner) \Rightarrow \overline{\text{Prov}_F(p, \ulcorner \neg A \urcorner)}.$$

Similarly, a formalized proof predicate is used to outline Hilbert’s Programme in (Kreisel 1968, p. 322).

Many of Kreisel's reflections on Hilbert's Programme, including the "amusing fact" mentioned above, are based on the technical properties of such a proof predicate. Thus, the question arises, how adequate is this formal rendering of Hilbert's ideas, which were rarely expressed in formal terms by Hilbert himself.

4 A Criticism of Kreisel's Criticism

While reading Kreisel, we identified—at least—three problems with his criticisms of Hilbert's Programme. First, his evaluation of this programme usually entangles positive and negative aspects which tarnish a fair assessment.¹⁵ Secondly, Kreisel concentrates too much on the consistency question which was even for Hilbert only a part of a "wider programme" (see Citation 19 below). Finally and, in our view, most problematically, Kreisel judges Hilbert's Programme from a blunt post-Gödelian perspective, i.e., with hindsight of the knowledge of Gödel's Incompleteness Theorems, including Gödel's formal apparatus.

4.1 A jumble of evaluations

In general, Kreisel offers a jumble of evaluations of Hilbert's programme, rather than a well-crafted philosophical assessment. In Section 3.2 we presented some evidence of positive remarks. However, let us take a closer look at Citation 4. There one finds compliments such as "[Hilbert's programme] was not an empty shell", "[it] led to precise progress", and "for some areas of informal mathematics... [it] can be carried out with striking results" alongside negative wordings as "[it] was refuted", "some of the uses... were ill thought out", and "even its success would not have provided...what Hilbert wanted". Furthermore, the paragraph contains negative catchwords which are explicitly *not* directed at Hilbert—namely "the kind of wise cracks which naive formalism invites"—but which easily can be confused by a "naive reader". As a matter of fact, it is quite common to accuse Hilbert of naive formalism and Kreisel was likely aware that his comments did little to correct this view.

In Citation 13 Kreisel lashes out at Brouwer *and* Hilbert, reproaching the latter for "logical atrocities with his claims for consistency". However, a clear argument explaining why the quest for a finitist consistency proof—which, by the way, is something else than a "claim for consistency", see the next subsection—should be considered an atrocity is lacking.

¹⁵ In fact of both are tarnished, Hilbert's Programme as well as Kreisel's evaluation of this Programme.

4.2 "Consistency is not the question"

First, let us comment on Kreisel's (*per se* not particularly original) observation that, in general, doubts about consistency are often more problematic than the consistency itself. This is an empirical fact that is difficult to deny. However Hilbert never expressed the slightest doubt about the consistency of analysis, let alone arithmetic.¹⁶

At best, Kreisel's observation would apply to intuitionists¹⁷ who are convinced of the inconsistency of classical mathematics. Probably, nobody seriously doubted the consistency of arithmetic, but the case might be different for analysis. It is possible to understand Hermann Weyl (1987, p. 1) expressing some doubts about the consistency of analysis, when he stated that "[the house of analysis] is to a large degree built on sand". And it is fair to say that even Gentzen (1969, p. 235)—who had some strong intuitionistic convictions, see (Gentzen 1937; Detlefsen 2015)—put such doubts on record in 1938:

Indeed, it seems not entirely unreasonable to me to suppose that *contradictions* might possibly be concealed even in classical *analysis*.

Still, there is nothing like that in Hilbert. As explained above, the modest goal of Hilbert's programme, to rebut intuitionism at its own game, does not start off with doubts on the consistency; quite the opposite. To put it polemically in terms of Kreisel's Citation 13: doubts about consistency are not part of "Hilbert's atrocities" but they are, at best, part of "intuitionist pollution". And when Kreisel points—quite correctly—in Citation 16 to the cumulative hierarchy of sets as the decisive concept to remove any remaining doubts

¹⁶ Set theory, however, might be a different case. Hilbert was indeed concerned with the paradoxes in naive set theory, see (Kahle 2006). But this was related to the—recognized—naivety of the first steps in this new area. After some reflection he saw a need, at best, for refinements, (Hilbert (1905), S. 204):

When I found it ["Hilbert's paradox", cf. (Peckhaus and Kahle 2002)], I first thought that it would put insurmountable obstacles in the way of set theory, on which it would have to fail; but now I believe certainly that, as always in science, after the revision of the foundations all essentials will survive.

Takeuti (Yasugi and Passell (2003), p. 122) puts this in the right perspective:

In the current day, axiomatic set theory is fully accepted and it is generally acknowledged that modern mathematics can be carried out in the framework of axiomatic set theory. No contradiction has arisen in axiomatic set theory, and a sense of security that no contradiction will arise in it in the future is supported by intuitive consensus. Under the current secure circumstances one cannot imagine the sense of crises of that earlier time.

¹⁷ It is worth adding, that in general intuitionists, in particular Brouwer, are not concerned with consistency but with meaning, see Footnote 7.

about consistency in Mathematics, he would have to put this argument forward in front of a skeptical intuitionist, not of Hilbert.¹⁸

We have argued in Section 2 that both Hilbert's Finitist Programme and its revised version must be understood within a broader context in which consistency is not the only objective. In fact, Kreisel put this observation already on record:

Citation 19 ((Kreisel 1983, p. 237)) *The quite fundamental, and generally neglected, distinction made here, between Hilbert's programme in the wide sense (adequacy of formal systems for representing reasoning) and in the narrow sense (consistency of particular systems)...*

When Kreisel insists on the question of consistency, he is merely targeting the "narrow Hilbert Programme", neglecting the "wide Hilbert Programme". It seems that Kreisel never questioned this wide programme, which encompasses all of his work in proof theory.¹⁹

Thus, the question arises as to the extent to which consistency is fundamental to proof theory. The author recalls two leading proof theorists of our time who, during an informal conversation repeatedly assured each other that, for modern proof theory: "Consistency is not the question!". Macintyre (2005, p. 2426) writes in this respect:²⁰

Much nonsense has been pronounced about Gentzen's work, even by extremely distinguished people. Consistency is not really the main issue at all. He did reveal fine structure in the unprovability of consistency of PA, as a consequence of much deeper general methodology.

The emphasis of consistency in his criticism of Hilbert²¹ is nearly reduced to absurdity when Kreisel himself recounts the following anecdote:²²

¹⁸ This is insofar unfair to Brouwer because, as mentioned in Footnotes 7 and 17, he was not interested in consistency. Ultimately, Kreisel's criticism of the quest for consistency misses the mark for both Hilbert and Brouwer. It would only undermine the supposed goal of Hilbert to beat Brouwer at his own game. However, it is different to claim that this game would be an "atrocious" than to say simply that it wasn't Brouwer's game.

¹⁹ Kreisel (1981, p. 85) explicitly eulogizes Hilbert's "*Tieferlegung der Fundamente*" (a term coined in (Hilbert 1918)), which we could identify as a *Leitmotiv* for the wider Programme.

²⁰ The fine structure mentioned in the citation is illustrated by a result about provably total functions of PA which one can obtain from Gentzen's work.

²¹ In contrast, Kreisel might correctly criticize others which would still today place consistency in the center of proof theory.

²² Just one page earlier, Kreisel emphasizes the importance of oral history:

Citation 20 ((Kreisel 2011, p. 43)) *According to Bernays [...] Hilbert was asked [...] if his claims for the ideal of consistency should be taken literally. In his (then) usual style, he laughed and quipped that the claims served only to attract the attention of mathematicians to the potential of proof theory.*

4.3 The post-Gödelian perspective

Our main criticism of Kreisel's critique of Hilbert's programme relates to his post-Gödelian perspective. In general, Hilbert's programme is evaluated with the knowledge of its failure, as if it would have been a *blind spot*²³ of Hilbert not to have seen this failure in the first place. Kreisel's evaluation always comes in hindsight with Gödel's results; in German it would be called: *historisierend*.²⁴

A characteristic example is given in Kreisel's presentation of *Hilbert's Second Problem* (Kreisel (1976a), p. 97). He first describes the ("Gödelian") effect of incompleteness that, for a statement $\phi(c_p)$ independent of a (consistent) arithmetical theory $\mathcal{F}$, the theory $\mathcal{F} + \{\neg\phi(c_p)\}$ is consistent, even if $\phi(c_p)$ is true over the integers. Then he states

Citation 21 ((Kreisel 1976a, p. 97)) *To use the language of Hilbert's second problem, even if the consistency of $\mathcal{F} + \{\neg\phi(c_p)\}$ ensures the existence of some concept satisfying all the theorems of this system, it does not ensure that the particular concept (of the natural numbers), for which axioms of arithmetic are intended, satisfies those theorems.*

But then he continues:

Citation 20a ((Kreisel 2011, p. 42)) For various reasons, I know more of Hilbert's nonmagisterial ideas; probably I am not conscious of most reasons, but one is that I saw a good deal of Bernays, who liked to remember Hilbert. (With such oral history in mind, published and semipublished material strikes the reader vividly that might otherwise be simply skipped. With the present industry of so-called history of logic or mathematics in bulky tomes, which also function as tombs for diverse bon mots, the chances are that hearsay (at least of — the likes of — Bernays) can also be documented). A most useful fact to remember is Hilbert's stroke in the mid-1920s, from which he never recovered. His style of thought was not the same after it.

For historical purposes, the reference to a stroke of Hilbert is interesting because no such event is mentioned in the comprehensively researched Hilbert biography by Constance Reid (1970). It seems, however, unlikely that Kreisel would have confused a stroke with the pernicious anemia from which Hilbert suffered in the mid-1920s.

²³ For this term see:

Citation 20b ((Kreisel 1980, p. 163)) Hilbert's conjectures [...] where refuted so simply that they have been described as 'blind spots'.

²⁴ In contrast to the literal English translation "historicizing", the German term implies that something is intentionally presented as historical in an artificial way.

Citation 22 (*Citation 21 continued*) *Hilbert's program is nothing else but this. For the formal systems of arithmetic $\mathcal{F}$...*

To give an f.c. [i.e., finitist] proof of the consistency of $\mathcal{F}$.

In this presentation, it seems that Hilbert would have neglected the philosophical implications of non-standard models of incomplete theories within the philosophical framework of his Programme. However, Hilbert did not think of incompleteness phenomena and undoubtedly always thought of complete theories. In particular, he would not have expected it already in a first-order theory. He had explicitly higher order theories in mind (see Hilbert and Ackermann 1928, opening section of Chapter 4). The non-axiomatizability of second-order logic, a corollary of Gödel's First Incompleteness Theorem, was, of course, unknown to Hilbert when he worked on his Programme.

More generally, however, the frequent use of a Gödelian proof predicate by Kreisel to represent Hilbert's ideas is a blunt historical distortion. Such a predicate was not on Hilbert's radar (and Kreisel knew it! See Citation 23 below).

According to Ernst Specker,²⁵ Finsler had said that it was of course known before Gödel that one could, in principle, encode derivations by natural numbers. However, it was clear that such an encoding could never be calculated in practice, so the method was considered useless. It was Gödel's brilliant observation that—for his “negative” result—it is unnecessary to *calculate* such codes. And Dana Scott confirmed this when he recalled Bernays answering his question of why the Hilbert school didn't consider Gödelization: They didn't see how to make use of it. In fact, Bernays (Hilbert and Bernays 1939, p. 207 (p. 216 of the second edition), footnote 1) put this on record as early as 1939:

The idea, to represent the symbolic formulas in this way by sequences of numerals and thereby to translate the rules for the formation of symbolic formulas and for the execution of formal proofs into arithmetical rules, was already conceived by HILBERT in connection with his reflections on Cantor's Continuumproblem. But at that time it appeared that the implementation of these ideas would become far too difficult.

Considering the vague nature of finitist mathematics into account, Kreisel was fully aware that Hilbert did not have any proof predicate for it:

Citation 23 ((Kreisel 1965, p. 177, footnote 44)) *Note that it was not clearly understood till [Gödel (1931)] that the then*

current finitistic proofs could be easily formalized in Z , for example, the use of nested recursion or equations involving higher type variables.

Moreover, he knew that Hilbert wouldn't even need such technicalities as long as he was working on a positive solution.²⁶

Citation 24 ((Kreisel 1965, p. 152)) *It is clear that there might be ambiguity in what exactly constitutes an elementary (consistency) proof. If one really expects to find one, one does not split hairs beforehand, but gives a precise formulation only after one has found such a proof. [...] Only when it comes to negative results is an analysis needed.*

Thus, Hilbert did not split hairs, but rather pursued *proof transformations*, mainly through the use of the ε -substitution method (Sieg 2011, §§ 4.1 and 4.2). Although Kreisel refers to the latter method in several places and was, notably, one of the few people outside the Hilbert school²⁷ to conduct research in the field of ε -substitution²⁸, he generally supersedes this method by speaking about proof predicates in his formal presentation of Hilbert's conception.

As we mentioned earlier, Bernays reported that Hilbert considered a provability predicate but found it too difficult. Thus, the key to Gödel's Theorems, as well as to later reflections on the logical complexity of such a predicate, as presented by Kreisel at various places, was not in Hilbert's hands. Therefore, it is unfair to build a historical understanding of proof theory on such a predicate. Even though Kreisel stresses that we only received such a provability predicate due to Hilbert work, putting the cart before the horse corrupts Hilbert's original conception.

5 Conclusion

Hilbert's Programme owes a lot to Kreisel. He (1958a) provided a first philosophical evaluation of it from a post-Gödelian perspective²⁹ and popularized the term. Later,

²⁶ In passing, we note that this observation contradicts the alleged “defect” stated in (1) of Citation 8.

²⁷ One can mention Ackermann and von Neumann as working with the ε -calculus; but even in the Hilbert school it became out of use quite soon.

²⁸ Delzell (1996, p. 114) writes: “It was not until the 50's that anyone applied the ε -theorems (...) to a known proof (of a non-universal, or even an interesting, theorem); that someone was G. Kreisel, beginning with his papers [(Kreisel 1951a, b)]...and [Kreisel (1958b)].”

²⁹ We did not intend to diminish Bernays presentation of Hilbert's Programme in (Hilbert and Bernays 1934, 1939). However, even though this presentation was written after Gödel's results became known, it is clearly still based on the original, pre-Gödelian conception of Hilbert.

²⁵ A story told in a lecture attended by the author of this paper.

with a “shift of emphasis”, he opened new avenues for proof-theoretic research, notably within mathematics.

In Section 3.2, we showed that Kreisel’s evaluation of Hilbert’s Programme is largely positive, contrary to the general impression one might have. This general impression is based on two different aspects.

First, it seems to have been an intrinsic part of Kreisel’s personality that he could not offer a positive evaluation without immediately adding a negative twist. Also, he cannot refrain from using bold language. In the second sentence of his obituary (Kreisel 1980) for Kurt Gödel the reader is confronted with “crackpots”, as if it were Gödel’s fault that some people have difficulties grasping the correct significance of his results. It is unfortunate that this attitude of Kreisel may overshadow the substance of his arguments, particularly for a reader “of a different species” as Kreisel (1987, pp. 161 and 177) would say.

Secondly, and more importantly, he questioned the philosophical rationale of Hilbert’s Programme. As we have argued, these doubts are mainly based on a *post-Gödelian perspective* and, therefore, cannot be directed at Hilbert. Kreisel expressed his repudiation of any form of a Revised Hilbert’s Programme by insulting colleagues working in this direction. It is worth noting, however, that he never made a negative remark about Bernays or his work—despite the fact that Bernays was the flagship of Hilbert’s Programme (in the original or any revised form).³⁰

In any case, criticizing Hilbert that his programme was ill thought out is akin to criticizing Columbus for starting from a wrong assumption when he set out to find a sea route to India. Today, proof theory is on a different shore than Hilbert had envisaged. It is on this shore that Kreisel went fishing with great success. In adapting his remark on Gödel given in Citation 6, we may say:³¹

The problems Kreisel worked on had been prepared and made accessible, mainly by Hilbert!

Funding Open Access funding enabled and organized by Projekt DEAL. Partial financial support was received from the Udo Keller Foundation and by national funds through the FCT—Fundação para a Ciência e a Tecnologia, I.P., under the scope of the projects UIDB/00297/2025 (<https://doi.org/10.54499/UIDB/00297/2025>) and UIDP/00297/2025 (<https://doi.org/10.54499/UIDP/00297/2025>)

³⁰ A colleague, who knew both, commented that Kreisel admired (in German: *verehrte*) Bernays.

³¹ One referee suggested that this thesis could serve as a starting point for a review of Kreisel’s technical work on extensions of Hilbert’s Programme. However, given the extent and diversity of Kreisel’s work, such a review would exceed the scope of this article. Regarding philosophical aspects in particular, we like to mention that Kreisel has proposed own ideas that go beyond Hilbert’s conception. For example, Kreisel’s notion of *Informal Rigour* (Kreisel 1967), see the excellent exposition in (Dean and Kurokawa 2026).

(Center for Mathematics and Applications).

Declarations

Competing interest The author has no relevant financial or non-financial interests to disclose.

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